



Dynamic ABC-Constrained Criterion Weighting for Strategic New Product Prioritization: A Fuzzy Decision-Analytics Approach

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Abstract: New product selection is a strategic decision problem in which market opportunities must be considered alongside financial requirements, technological capabilities, and production constraints. Existing fuzzy multi-criteria decision-making (MCDM) methods can rank product alternatives under uncertainty, but they do not always maintain consistency between the priority classes assigned to evaluation criteria and their final weights. This study develops a fuzzy decision-analytics approach that links Pareto-based ABC classification with dynamic criterion weighting for strategic new product prioritization. Linguistic assessments from nine experts were represented by triangular fuzzy numbers (TFNs) and combined with quantitative data to evaluate eight candidate products against ten market, economic, technological, and production-related criteria. The criteria were first assigned to ABC classes, after which class-specific scaling factors were calculated to preserve their relative importance within each class while enforcing the required weight order across classes. The resulting weights were then used to rank and classify the candidate products. The procedure maintained the specified priority structure among the criteria. Transport racks obtained the highest overall decision value of 0.825 and were assigned to Class A, followed by excavator buckets with a value of 0.769 and excavator chassis with a value of 0.704. Six products were placed in Class B, while quick couplers ranked last with a value of 0.611 and were assigned to Class C. The results indicate that integrating ABC classification directly into criterion weighting provides a consistent basis for product ranking and priority grouping. The proposed approach supports manufacturing managers in aligning new product decisions with strategic objectives, existing production capabilities, and resource constraints.

Keywords: Strategic new product selection; Decision analytics; Fuzzy multi-criteria decision-making; Dynamic criterion weighting; ABC classification; Manufacturing strategy

1 Introduction

The introduction of a new product into a company’s production program is a decision-making problem involving different business functions within a company. Moreover, it is not a problem addressed only at the strategic level, as managers at the operational and tactical levels are also involved in this process. Such a decision depends on numerous factors, and therefore both economic indicators and the technical and market requirements of new products need to be analyzed. Precisely because a large number of factors are considered simultaneously, often involving conflicting requirements, new product selection represents a complex and challenging multi-criteria decision-making (MCDM) problem. In this regard, Mousavi et al. [1] emphasize that the initial choices can have a significant impact on a company’s economic and market success. In addition, some studies have used a similar approach, based on fuzzy sets and multi-attribute techniques, in new product development (NPD) and introduction [2, 3]. Today, in a period of rapid innovation and increasing competition in local and global markets, companies rely on MCDM even more to screen ideas and address early-stage uncertainty [4–8].

Li et al. [4], by combining fuzzy set theory with the Best-Worst Method (BWM) [9], evaluated new product ideas across five core perspectives: financial, marketing, engineering, manufacturing, and sustainability. Other sector-specific applications followed. Chen et al. [5] developed an MCDM framework for the OLED industry, while Göçer and Büyükoçkan [6] used a different approach for ranking competing NPD alternatives, applying a Pythagorean fuzzy Multi-Objective Optimization by Ratio Analysis (MULTIMOORA) model [10] to prioritize them. Different

priorities often conflict in business decisions. In such cases, MCDM methods offer practical tools for making trade-offs.

Expert judgments by their nature involve subjectivity and vagueness, which make fuzzy approaches a suitable tool for expressing assessments in different types of problems, including new product selection. The basic concept of fuzzy sets was introduced by Zadeh [11] back in 1965, while Bellman and Zadeh [12] adapted the concept itself and made it suitable for application in MCDM problems. The basic idea behind the concept was that experts, in cases where something cannot be assessed using precise numerical values, can use linguistic expressions that are then modeled using fuzzy numbers. It is precisely this advantage, namely the possibility of making assessments while accounting for a certain degree of uncertainty, that led to wide adoption across complex decision problems, as Mousavi et al. [1] showed early on in group decision-making for new product selection. In addition, Sumrit integrated fuzzy Step-wise Weight Assessment Ratio Analysis (SWARA) [13] and fuzzy Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [14] to evaluate supplier co-design capabilities [7], while Mousavi et al. analyzed risks associated with competitors' cooperation in NPD projects [8]. Furthermore, Park et al. provided broader evidence by surveying 266 studies on the “fuzzy front end” of NPD [15]. This study highlighted the great importance of systematic approaches in which decisions are based on qualitative, incomplete, or subjective information.

Categorization is frequently performed in the relevant literature using ABC analysis [16]. The method is basically based on the Pareto principle, classifying elements according to their importance into three groups: A, B, and C. Traditional ABC setups, however, evaluate only one criterion at a time, a limitation that drastically reduces their real-world utility. That gap drove the move toward multi-criteria ABC approaches. In paper [16], the authors first introduced the use of multiple criteria in ABC classification, and later work added optimization and MCDM techniques [17–19]. Research continues to expand these models into multi-criteria and uncertain decision environments. Yiğit and Esnaf [20] built a three-phase system merging analytic hierarchy process (AHP) [21, 22], fuzzy C-means, and a modified veto rule for multi-criteria ABC inventory classification, while maintaining compliance with the Pareto principle. Yung et al. [23] designed a multi-attribute fuzzy ABC classification system for space-mission components, showing how qualitative and quantitative information can be incorporated within the classification process. Moving further into uncertainty, Razavi Hajiagha et al. [24] combined fuzzy and stochastic information within a multi-criteria ABC classification framework. Together, these studies confirm an active shift from single-criterion ABC tools toward multi-criteria fuzzy approaches.

Alternative computational methods have also reshaped multi-criteria ABC classification. Tavassoli and Farzipoor Saen [25] developed a stochastic data envelopment analysis (DEA) [26] approach for inventory sorting under uncertainty, while Saracoglu [27] developed a scatter-search algorithm designed for addressing a similar classification problem. In addition, the classification problem was also considered [28], where the authors developed and tested an explainable artificial intelligence framework for multi-criteria ABC classification to address the “black box” interpretability problem and clarify how items are classified into specific groups. Paredes Rodríguez et al. [29] took a hybrid route, combining fuzzy AHP and TOPSIS for ABC classification based on both qualitative and quantitative criteria. More recently, researchers turned their attention to the criteria-selection step itself. As highlighted by Theunissen et al. [30], the way criteria are defined and weighted can substantially affect the resulting inventory classification. Based on the above, it can be concluded that contemporary ABC approaches are used across different methodological domains, ranging from multi-criteria evaluation and uncertainty to advanced computational methods and model interpretability.

Although multi-criteria ABC is frequently used in the literature, the problem of product selection has not often been addressed using this approach. Fuzzy MCDM approaches for new product selection and multi-criteria ABC approaches have largely developed as separate research streams. New product selection literature concentrates almost entirely on weighting and ranking alternatives under uncertainty, whereas ABC research mainly addresses the classification of inventory items. Modern fuzzy MCDM studies demonstrate sophisticated ranking tools for new product ideas and NPD-related alternatives; ABC studies rely on fuzzy clustering, stochastic modeling, DEA, TOPSIS, Tomada de Decisão Interativa e Multicritério (TODIM), and explainable machine-learning techniques for inventory classification. However, these approaches do not explicitly address the dynamic adjustment of criterion weights according to the resulting ABC classes. This may lead to an inconsistency in which a criterion assigned to a lower-priority class receives a higher final weight than a criterion belonging to a higher-priority class. As a consequence, the ABC classification may not be properly reflected in the final weighting structure.

This issue becomes particularly relevant when ABC classification is applied to the criteria themselves rather than to alternatives. Categorizing criteria establishes a clear hierarchy of importance. Criteria in class A should carry greater weight than those in class B, while class B criteria should have greater weight than those in class C. Standard normalization of initial criterion weights, however, often fails to preserve this hierarchy. Existing multi-criteria ABC models focus almost entirely on assigning alternatives or inventory items to classes. The interaction between ABC-based criterion classification and subsequent weight adjustments receives considerably less attention. This gap calls for a clear mechanism to translate ABC classification results into a consistent final weighting structure.

To address this gap, a fuzzy ABC-based approach was developed in this study for new product prioritization and selection. The approach combines fuzzy evaluation with ABC classification and is based on the principle of dynamic criterion weighting, where the initial weights of criteria are adjusted according to the classes to which they are assigned. The resulting criterion weights are then used as inputs to the second phase of the model, which involves prioritizing candidate products. This mechanism preserves the relative importance of criteria within each class while enforcing the required hierarchy between classes ($A \geq B \geq C$). In this way, ABC classification moves from a simple sorting step to an integral part of the criterion-weighting process.

The main objective is to develop an integrated fuzzy MCDM and ABC methodology that keeps criterion classification aligned with final criterion weights. The proposed approach is demonstrated through a new product selection case and provides a transparent and consistent prioritization framework. The primary contribution is the dynamic criterion-weighting mechanism directly linked to ABC classes, i.e., a methodological link that is not explicitly addressed in existing fuzzy MCDM and multi-criteria ABC approaches.

2 Conceptual Framework of the Fuzzy ABC Approach

This section outlines the methodological framework designed to select and prioritize new products within an existing manufacturing program. Called the fuzzy ABC approach, the model combines expert linguistic judgments, represented by triangular fuzzy numbers (TFNs), with a Pareto-based ABC classification mechanism. The evaluation runs through two stages. First, the model determines criterion weights and their relative importance. Next, it evaluates, ranks, and classifies candidate products according to their overall decision performance.

2.1 Problem Formulation and Decision-Making Framework

The considered problem is formulated as an MCDM problem in which a decision maker or a group of experts evaluates a set of candidate products according to a predefined set of criteria. Let $i, i = 1, \dots, I$ denote the candidate products and $j, j = 1, \dots, J$ the evaluation criteria. The performance of the candidate products is represented by the fuzzy decision matrix $[\tilde{x}_{ij}]_{I \times J}$, where $\tilde{x}_{ij}$ denotes the performance of alternative i with respect to criterion j . The criteria may be quantitative or qualitative and can be classified as benefit or cost criteria according to their preferred direction of performance.

Since the relative importance of the criteria is not known a priori, it is determined as an integral part of the proposed fuzzy ABC approach. Experts assess the importance of the criteria using linguistic terms represented by TFNs. The aggregated assessments are then processed according to the Pareto principle to determine the relative contribution of each criterion, classify the criteria into A, B and C classes, and derive their final weights.

The obtained criterion weights are subsequently used to evaluate the candidate products. After normalization and aggregation of the criterion-specific performance values, an overall value is obtained for each alternative. The alternatives are then ranked according to their relative contribution and classified into A, B and C priority classes using the same Pareto-based principle.

Thus, the proposed approach provides both the criterion weights and the final ranking and ABC classification of the candidate products, enabling a transparent identification of the alternatives with the highest decision priority.

2.2 Linguistic Evaluation and Fuzzy Representation

The evaluation of candidate products involves both quantitative and qualitative information. While some criteria can be expressed numerically, others, such as market demand, technological feasibility, production capacity compatibility and market potential, are inherently subjective and difficult to quantify precisely. Therefore, the proposed fuzzy ABC approach employs linguistic evaluations to capture expert judgments in a form that better reflects the uncertainty and imprecision of such assessments.

The evaluation is performed by a multidisciplinary team of nine experts, each assessing the importance of the evaluation criteria and the performance of candidate products according to the relevant criteria. The individual judgments are subsequently aggregated to obtain collective assessments. The linguistic evaluations are represented using TFNs [31–33] defined by three values representing the lower, modal and upper values of an assessment.

The linguistic scale is applied in both stages of the proposed approach. First, it is used to assess the relative importance of the evaluation criteria, which subsequently serves as the basis for their fuzzy ABC weighting. Second, it is used to evaluate candidate products with respect to qualitative criteria for which reliable numerical measurements are not available.

The aggregated fuzzy assessments are subsequently converted into representative numerical values for further processing within the fuzzy ABC procedure.

2.3 Pareto-Based ABC Classification Principle

The proposed fuzzy ABC approach is divided into two stages: first, to classify the evaluation criteria according to their relative contribution to the total criterion importance, and second, to classify the candidate products according to

their relative contribution to the overall decision value.

After calculating the relative contribution of each element, the elements are arranged in descending order and their cumulative contribution is determined. Based on the cumulative contribution, three priority classes are defined. Class A includes the elements with the highest contribution, Class B includes elements with an intermediate contribution, while Class C includes the remaining elements, which have a very low contribution.

In accordance with the adopted Pareto classification, Class A covers approximately the first 20% of the cumulative contribution, Class B extends from the end of Class A to approximately 95%, and the remaining elements are assigned to Class C. The same classification principle is applied consistently to both criteria and candidate products.

This mechanism enables the proposed approach to provide not only a numerical ranking but also a meaningful classification of criteria and alternatives according to their relative contribution to the decision-making process.

2.4 Conceptual Structure of the Fuzzy ABC Approach

As already stated, the proposed fuzzy ABC approach integrates fuzzy expert evaluation with Pareto-based ABC classification to determine criterion weights and prioritize candidate products. The approach consists of two sequential stages (Figure 1).

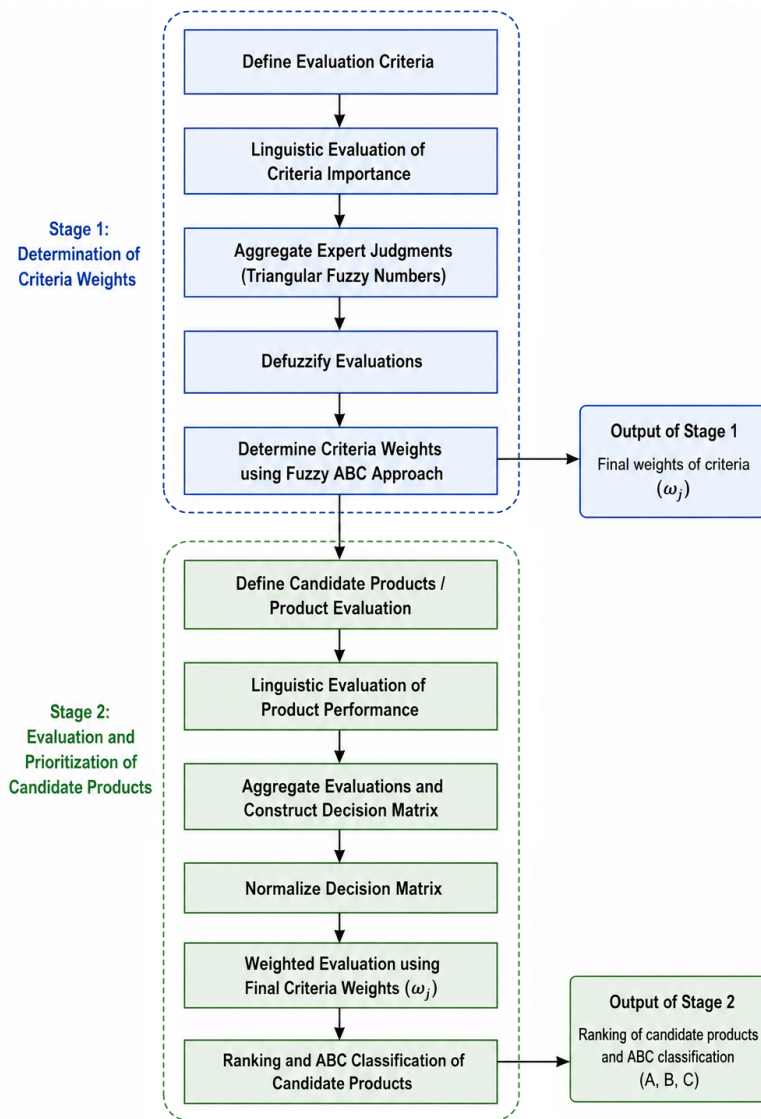


Figure 1. Conceptual structure of the proposed fuzzy ABC approach

Note: ABC refers to the Pareto classification of candidate products into Classes A, B, and C.

In the second stage, the obtained criterion weights are applied to the normalized evaluation of candidate products. Their overall values and relative contributions are determined, followed by ranking and ABC classification according to the Pareto principle.

The detailed mathematical formulation of both stages is presented in Section 3.

3 Mathematical Formulation of the Fuzzy ABC Approach

This section presents the mathematical formulation of the proposed fuzzy ABC approach introduced in the previous section. The formulation covers the representation and aggregation of fuzzy expert assessments, defuzzification, determination of criterion weights based on expert assessments, normalization of the decision matrix, and prioritization of candidate products.

3.1 Triangular Fuzzy Numbers and Expert Judgment Aggregation

TFNs are used to represent the linguistic assessments provided by the experts. A TFN is defined as:

$$\tilde{x} = (x^l, x^m, x^u) \quad (1)$$

where, x^l , x^m , and x^u denote the lower, modal, and upper values, respectively.

Five linguistic terms are used to express the expert assessments: very low value (V1), low value (V2), medium value (V3), high value (V4), and very high value (V5). Their corresponding TFN representations are defined on a 1–10 scale and are presented in Table 1.

Table 1. Linguistic terms and corresponding triangular fuzzy numbers (TFNs)

Linguistic Term (Benefit Criteria)	Code (Benefit Criteria)	Linguistic Term (Cost Criteria)	Code (Cost Criteria)	TFN
Very low value	V1	Very high value	V5	(1, 1, 3.5)
Low value	V2	High value	V4	(2, 3.5, 5)
Medium value	V3	Medium value	V3	(4, 5.5, 7)
High value	V4	Low value	V2	(6, 7.5, 9)
Very high value	V5	Very low value	V1	(7.5, 10, 10)

The aggregation of expert assessments follows the standard component-wise arithmetic operations for TFNs [31–33]. Thus, the lower, modal, and upper values of the TFNs are aggregated separately across the expert panel.

The same panel of E experts provided two types of assessments within the decision-making process. First, the experts assessed the relative importance of each evaluation criterion. Second, they evaluated the performance of each candidate product with respect to each criterion. For the assessment of criterion importance, the aggregated TFN for criterion j is calculated as:

$$\tilde{w}_j = \left(\frac{1}{E} \sum_{e=1}^E w_e^l, \frac{1}{E} \sum_{e=1}^E w_e^m, \frac{1}{E} \sum_{e=1}^E w_e^u \right) \quad (2)$$

For the qualitative evaluation of candidate products, the aggregated fuzzy assessment of product i with respect to criterion j is calculated as:

$$\tilde{x}_{ij} = \left(\frac{1}{E} \sum_{e=1}^E x_e^l, \frac{1}{E} \sum_{e=1}^E x_e^m, \frac{1}{E} \sum_{e=1}^E x_e^u \right) \quad (3)$$

where, $e = 1, \dots, E$, denotes the expert index and E is the total number of experts. Here w_e^l , w_e^m , and w_e^u denote the lower, modal, and upper values of the assessment provided by expert e , for the considered criterion.

The aggregated fuzzy criterion-importance values and product evaluations are subsequently defuzzified using the centroid method [34–36]. The crisp importance value of criterion j is calculated as:

$$w_j = \frac{w_j^l + w_j^m + w_j^u}{3} \quad (4)$$

where, w_j^l , w_j^m , and w_j^u denote the lower, modal, and upper values of the aggregated fuzzy importance of criterion j , respectively.

The crisp evaluation of candidate product i with respect to criterion j is calculated as:

$$x_{ij} = \frac{x_{ij}^l + x_{ij}^m + x_{ij}^u}{3} \quad (5)$$

where, x_{ij}^l , x_{ij}^m , and x_{ij}^u denote the lower, modal, and upper values of the aggregated fuzzy value of candidate product i , respectively.

The resulting crisp criterion-importance values are used in the fuzzy ABC weighting procedure, whereas the defuzzified qualitative evaluations, together with the exact quantitative data, form the crisp decision matrix used for subsequent normalization and product prioritization.

3.2 Normalization of the Decision Matrix

The crisp decision matrix $[x_{ij}]_{I \times J}$ combines exact quantitative data with defuzzified values obtained from aggregated fuzzy assessments for qualitative criteria. Since the criteria may have different units and preferred directions of performance, normalization is performed to transform their values into a comparable scale.

For benefit criteria, where higher values are preferred, the normalized value is calculated as:

$$n_{ij} = \frac{x_{ij}}{\max_i x_{ij}} \quad (6)$$

For cost criteria, where lower values are preferred, the normalized value is calculated as:

$$n_{ij} = \frac{\min_i x_{ij}}{x_{ij}} \quad (7)$$

The resulting normalized decision matrix is defined as:

$$[n_{ij}]_{I \times J} \quad (8)$$

where, $n_{ij} \in [0, 1]$. Thus, higher normalized values represent more desirable performance for both benefit and cost criteria.

The values of the normalized decision matrix are weighted using the criterion weights determined in the previous stage. In this way, the performance values and overall values of the candidate products are calculated.

3.3 Fuzzy ABC-Based Weighting of Criteria

The relative importance of the evaluation criteria is determined independently of the candidate-product performance evaluations. The defuzzified criterion-importance values w_j , obtained from the aggregated expert assessments, are first converted into relative contributions:

$$f_j = \frac{w_j}{\sum_{j=1}^J w_j} \quad (9)$$

The criteria are sorted according to their f_j values in descending order. The cumulative contribution is subsequently calculated as:

$$cf_j = \sum_{h=1}^j f_h \quad (10)$$

where, the index h refers to the position of a criterion in the sorted sequence.

Based on the cumulative contribution, the criteria are classified into three Pareto classes. Class A comprises the criteria contributing to approximately the first 20% of the cumulative contribution, Class B covers the subsequent range up to approximately 95%, and the remaining criteria are assigned to Class C.

The Pareto classification determines the membership of criteria in Classes A, B and C. To ensure consistency with the ABC hierarchy, the preliminary criterion weights are required to satisfy the condition that every criterion belonging to a higher-priority class has a weight greater than or equal to that of every criterion belonging to a lower-priority class. Accordingly:

$$\min_{j \in A} p_j \geq \max_{j \in B} p_j \quad (11)$$

and

$$\min_{j \in B} p_j \geq \max_{j \in C} p_j \quad (12)$$

To preserve the relative contribution of criteria within each class, a class-specific scaling factor s_g is assigned to each ABC class. The preliminary weight of criterion j is calculated as:

$$p_j = s_g f_j, \quad j \in g, \quad g \in \{A, B, C\} \quad (13)$$

The scaling factors are determined sequentially, starting from Class C. The scaling factor of Class C is set to $s_C = 1$, while the factors for Classes B and A are determined to satisfy the class-order constraints with the minimum required scaling:

$$s_C = 1 \quad (14)$$

$$s_B = \frac{\max_{j \in C} f_j}{\min_{j \in B} f_j} \quad (15)$$

$$s_A = \frac{s_B \max_{j \in B} f_j}{\min_{j \in A} f_j} \quad (16)$$

This mechanism ensures that a criterion with the lowest weight within one class (A or B) has a weight greater than or equal to that of a criterion with the highest weight within the lower class (B or C). In this way, a clear distinction between the classes is established, preventing a criterion belonging to a lower-priority class from receiving a higher weight than a criterion from a higher-priority class, which would be methodologically unjustified.

The final criterion weight is then obtained by normalization:

$$\omega_j = \frac{p_j}{\sum_{j=1}^J p_j} \quad (17)$$

Since all preliminary weights are normalized by the same denominator, the resulting final criterion weights preserve the ABC hierarchy. Consequently, they satisfy:

$$\min_{j \in A} \omega_j \geq \max_{j \in B} \omega_j \quad (18)$$

and

$$\min_{j \in B} \omega_j \geq \max_{j \in C} \omega_j \quad (19)$$

Consequently, the final criterion weights satisfy both the ABC hierarchy and the normalization condition:

$$\sum_{j=1}^J \omega_j = 1 \quad (20)$$

This procedure prevents a criterion belonging to a lower-priority class from receiving a higher weight than a criterion belonging to a higher-priority class solely due to differences in class size. At the same time, the relative contribution of criteria within each class is retained.

The obtained vector of criterion weights is subsequently used in the evaluation and prioritization of the candidate products.

3.4 Fuzzy ABC-Based Product Prioritization

The obtained criterion weights are incorporated into the normalized decision matrix to determine the weighted normalized performance of each candidate product. For each candidate product i and criterion j , the weighted normalized value is calculated as:

$$z_{ij} = n_{ij} \omega_j \quad (21)$$

The overall value of candidate product i is then determined by aggregating its weighted normalized performances:

$$r_i = \sum_{j=1}^J z_{ij} \quad (22)$$

To express the relative contribution of each candidate product to the total decision value, the normalized individual value is calculated as:

$$f_i = \frac{r_i}{\sum_{i=1}^I r_i} \quad (23)$$

The candidate products are sorted according to their f_i values in descending order. The cumulative contribution is subsequently calculated as:

$$cf_i = \sum_{h=1}^i f_h \quad (24)$$

where, h denotes the position of a candidate product in the sorted sequence.

Based on the cumulative contribution, the candidate products are classified into three Pareto classes. Class A comprises the products contributing to approximately the first 20% of the cumulative contribution, Class B covers the subsequent range up to approximately 95%, and the remaining products are assigned to Class C.

The resulting order of f_i values represents the final ranking of the candidate products, while the ABC classification indicates their relative priority.

4 Case Study

This section presents the application of the proposed fuzzy ABC approach to a practical case study. As previously explained, the methodology is designed to prioritize candidate products that may potentially be included in a company's manufacturing program. The case study was conducted in a real-world decision-making environment involving a multidisciplinary expert team, which is presented in the following section.

4.1 Company and Decision Context

The case study considers a manufacturing company specializing in the design and production of welded steel structures and components for construction machinery. The company's activities include material preparation, computer numerical control (CNC) machining, manual and robotic welding, surface protection, assembly and final quality control. The company employs approximately 500 workers, including around 30 engineers from different technical fields, and its production system supports both individual and serial production of complex welded structures.

The company's main customers are manufacturers of construction, industrial and heavy machinery, primarily on the European market, while part of its production is also intended for the spare-parts market. Its strategic orientation is focused on developing higher value-added products, increasing exports and continuously expanding the production program in accordance with market requirements.

Within its five-year strategic development plan, the company identified the need to expand its production program in order to reduce dependence on a limited number of customers, increase export potential and improve the utilization of existing production capacities. Consequently, a systematic product selection process was initiated based on market conditions, customer requirements, technological capabilities and the company's strategic objectives.

For this purpose, the company's management formed a multidisciplinary engineering team that identified potential products that could be included in the company's manufacturing program and that are the subject of analysis in this study.

4.2 Candidate Products

Based on the market analysis, customer requirements, technological capabilities and strategic objectives of the company, eight candidate products were selected for evaluation. The selected products belong to the field of construction machinery and welded steel structures and differ in their technical complexity, required investment, production requirements and market potential.

The considered set of candidate products consists of pallet forks ($i = 1$), excavator bucket ($i = 2$), excavator arm ($i = 3$), dumper trough ($i = 4$), modular steel bridge ($i = 5$), quick couplers ($i = 6$), transport racks ($i = 7$), and excavator chassis ($i = 8$).

These products were selected because they are compatible with the company's existing production technologies while exhibiting different levels of market potential, investment requirements and production complexity. This diversity makes them suitable for evaluating the proposed fuzzy ABC approach for new-product selection.

4.3 Evaluation Criteria

The evaluation criteria were defined by the expert team to capture the main market, economic, technological and production-related aspects affecting the successful introduction of a new product. The criteria were selected considering the company's strategic objectives, existing production capabilities and the characteristics of the candidate products.

The following 10 evaluation criteria are considered: market demand ($j = 1$), required investment ($j = 2$), technological feasibility ($j = 3$), expected profitability ($j = 4$), production implementation time ($j = 5$), compatibility with production capacities ($j = 6$), time to market ($j = 7$), export potential ($j = 8$), production complexity ($j = 9$), and market potential ($j = 10$).

Market demand represents the expected demand for the considered product in the target market. Required investment refers to the financial resources needed for product development and introduction (expressed in Euro (EUR)). Technological feasibility refers to the assessment of the extent to which the existing technology within the company is suitable for manufacturing the product under consideration, i.e., whether and to what extent the technological requirements are met. Expected profitability represents the expected average annual profit from the sale of the product under consideration, expressed in EUR/year.

Production implementation time refers to the time required to prepare the product for serial production and is expressed in months. Compatibility with production capacities reflects the extent to which the product can be manufactured using existing equipment. Time to market represents the time required from the development decision to commercialization (expressed in months). Export potential reflects the possibility of placing the product on international markets. Production complexity represents the complexity of the required production process. Since lower production complexity is preferred, this criterion is treated as a cost criterion in the evaluation. Finally, market potential reflects the size and breadth of the potential market.

4.4 Expert Team and Data Collection

The evaluation was performed by a multidisciplinary expert team consisting of nine experts with complementary expertise in product development, design, engineering analysis, manufacturing, production, sales and marketing, and finance. The team included a research and development (R&D) manager ($e = 1$), lead design engineer ($e = 2$), senior design engineer ($e = 3$), computer-aided engineering/finite element method (CAE/FEM) analysis engineer ($e = 4$), manufacturing engineer ($e = 5$), welding engineer ($e = 6$), production manager ($e = 7$), sales and marketing manager ($e = 8$), and finance manager ($e = 9$).

Each expert assessed the importance of the ten evaluation criteria and evaluated the performance of the eight candidate products with respect to qualitative criteria. The assessments were expressed using the five-level linguistic scale defined in Table 1, ranging from very low value (V1) to very high value (V5). The resulting aggregated expert assessments form the basis for the fuzzy decision matrix and criteria prioritization.

4.5 Results of the Fuzzy ABC Approach

In accordance with the mathematical model presented in Section 3, the evaluation process was executed through a structured computation. This dynamic scaling approach independently assesses the criteria importance while strictly maintaining the hierarchy between Pareto classes.

4.5.1 Evaluation of criteria importance and weighting

Following the evaluation principle outlined in Section 3.3, the expert panel assessed the relative importance of the ten criteria using linguistic variables. The aggregated and defuzzified criterion-importance values w_j were computed using Eqs. (2) and (4) and subsequently converted into relative contributions f_j to form the basis for ABC classification (Eq. (9)).

To ensure that no criterion from a lower-priority class overrides a criterion from a higher-priority class, the ABC classification and dynamic scaling procedure (Eqs. (10)–(20)) was applied. Based on the class-boundary constraints, the scaling factors were calculated as $s_C = 1$, $s_B = 0.818$, and $s_A = 0.710$. These factors generated preliminary weights p_j , which were then normalized into final criteria weights ω_j . Table 2 details the linguistic assessments, intermediary parameters, and final weights.

Table 2. Linguistic assessment, defuzzified importance, and final weighting of criteria

j	Expert Judgments	w_j	f_j	cf_j	ABC Class	p_j	ω_j
$j = 2$	V2, V5, V5, V5, V5, V5, V5, V5, V5	8.537	0.155	0.155	A	0.11	0.14
$j = 1$	V3, V3, V3, V3, V5, V5, V5, V5, V5	7.537	0.137	0.293	B	0.097	0.12
$j = 4$	V1, V3, V3, V3, V3, V4, V5, V5, V5	6.537	0.119	0.412	B	0.097	0.12
$j = 3$	V1, V1, V1, V4, V4, V4, V4, V4, V4	5.611	0.102	0.514	B	0.084	0.11
$j = 6$	V1, V1, V2, V2, V4, V4, V4, V4, V5	5.537	0.101	0.615	B	0.083	0.10
$j = 8$	V2, V2, V2, V2, V2, V2, V5, V5, V5	5.389	0.098	0.713	B	0.08	0.10
$j = 9$	V1, V2, V2, V2, V3, V4, V4, V4, V4	5.315	0.097	0.810	B	0.079	0.10
$j = 5$	V2, V2, V2, V2, V2, V2, V2, V3, V4	4.167	0.076	0.885	B	0.062	0.08
$j = 7$	V1, V1, V1, V1, V1, V2, V3, V3, V4	3.463	0.063	0.948	B	0.052	0.07
$j = 10$	V1, V1, V1, V1, V1, V1, V2, V2, V4	2.833	0.052	1.000	C	0.052	0.06

Note: j denotes the criterion index; w_j denotes the defuzzified importance value of criterion j ; f_j denotes its relative contribution; cf_j denotes its cumulative contribution; p_j denotes its preliminary weight; and ω_j denotes its final normalized weight.

The application of class-specific scaling factors successfully harmonized the transition between classes. The minimum preliminary weight in Class A ($p_2 = 0.11$) perfectly equilibrates with the maximum in Class B ($p_1 = 0.097$), just as the minimum in Class B ($p_7 = 0.052$) directly matches the maximum in Class C ($p_{10} = 0.052$). This rigorously preserves the ABC hierarchy.

4.5.2 Candidate product evaluation and decision matrix normalization

The product evaluation matrix combines exact quantitative data ($j = 2, j = 4, j = 5$, and $j = 7$) with defuzzified qualitative assessments ($j = 1, j = 3, j = 6, j = 8, j = 9$, and $j = 10$). Standard V1–V5 coding was applied for benefit criteria, while inverse linguistic coding was applied to Production complexity ($j = 9$) to ensure that higher crisp values correspond to lower production complexity. Accordingly, benefit-type normalization was applied to the transformed values of this criterion. The complete initial decision matrix is presented in Table 3.

To ensure scale comparability, the initial decision matrix was normalized using Eqs. (6) and (7) for benefit and cost criteria, respectively. The resulting normalized decision matrix is shown in Table 4.

Table 3. Initial crisp decision matrix

Product	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j = 7$	$j = 8$	$j = 9$	$j = 10$
$i = 1$	6.500	1,000,000	4.907	600,000	8	4.167	10	7.130	5.426	6.537
$i = 2$	6.611	500,000	6.722	300,000	4	7.278	6	6.500	4.87	5.907
$i = 3$	6.315	700,000	5.500	420,000	9	5.685	12	6.463	6.056	7.352
$i = 4$	4.870	800,000	6.574	480,000	10	7.574	10	6.278	4.907	4.981
$i = 5$	5.722	600,000	6.241	360,000	12	6.426	16	6.315	7.167	5.722
$i = 6$	5.278	1,000,000	5.093	600,000	16	4.204	18	6.944	6.796	5.463
$i = 7$	5.278	250,000	7.537	150,000	3	7.500	4	5.463	2.019	7.130
$i = 8$	5.907	350,000	6.278	210,000	6	5.796	6	7.241	4.574	4.426

Table 4. Normalized decision matrix

Product	$j = 1$	$j = 2$	$j = 3$	$j = 4$	$j = 5$	$j = 6$	$j = 7$	$j = 8$	$j = 9$	$j = 10$
$i = 1$	0.983	0.250	0.651	1.000	0.375	0.550	0.400	0.985	0.757	0.889
$i = 2$	1.000	0.500	0.892	0.500	0.750	0.961	0.667	0.898	0.680	0.804
$i = 3$	0.955	0.357	0.730	0.700	0.333	0.751	0.333	0.893	0.845	1.000
$i = 4$	0.737	0.312	0.872	0.800	0.300	1.000	0.400	0.867	0.685	0.678
$i = 5$	0.866	0.417	0.828	0.600	0.250	0.848	0.250	0.872	1.000	0.778
$i = 6$	0.798	0.250	0.676	1.000	0.188	0.555	0.222	0.959	0.948	0.743
$i = 7$	0.798	1.000	1.000	0.250	1.000	0.990	1.000	0.754	0.282	0.970
$i = 8$	0.894	0.714	0.833	0.350	0.500	0.765	0.667	1.000	0.638	0.602

4.5.3 Final prioritization of candidate products

By applying the proposed dynamic scaling procedure, the relative individual contribution and cumulative contribution were subsequently calculated using Eq. (21) to Eq. (24). The alternatives were then sorted in descending order based on their overall values, and ABC classification was applied, as presented in Table 5.

Table 5. Final ranking and ABC prioritization of candidate products

Product	r_i	f_i	cf_i	ABC Class	Rank
Transport racks ($i = 7$)	0.825	0.148	0.148	A	1
Excavator bucket ($i = 2$)	0.769	0.138	0.286	B	2
Excavator chassis ($i = 8$)	0.704	0.127	0.413	B	3
Excavator arm ($i = 3$)	0.672	0.121	0.534	B	4
Pallet forks ($i = 1$)	0.669	0.120	0.654	B	5
Dumper trough ($i = 4$)	0.659	0.118	0.772	B	6
Modular steel bridge ($i = 5$)	0.655	0.118	0.890	B	7
Quick couplers ($i = 6$)	0.611	0.110	1.000	C	8

Note: i denotes the candidate product index; r_i denotes the overall value of candidate product i ; f_i denotes its relative contribution; and cf_i denotes its cumulative contribution.

4.6 Discussion of Results

The final ranking is presented in Table 5 and is based on the overall decision values, which combine the weighted and normalized performance of the candidate products across all evaluation criteria.

Transport racks achieved the highest overall decision value of 0.825, with a relative contribution of 0.148, and was ranked first among the eight candidate products. Its position results from the combined evaluations across all considered criteria and the final criterion weights. In particular, favorable performance with respect to several highly weighted criteria contributed to its first-place ranking.

Excavator bucket was ranked second, with an overall decision value of 0.769 and a relative contribution of 0.138. It was followed by excavator chassis and excavator arm, which ranked third and fourth, with overall decision values of 0.704 and 0.672, respectively.

Pallet forks, dumper trough, modular steel bridge, and quick couplers occupied the fifth to eighth positions, respectively. Their overall decision values ranged from 0.611 to 0.669. Compared with the higher-ranked products,

these alternatives obtained lower aggregated values based on the considered market, economic, technological, and production-related criteria.

The ABC classification provides an additional view of the obtained ranking. Transport racks were assigned to Class A and therefore represent the highest-priority product for further consideration. The remaining products, except quick couplers, were assigned to Class B. Quick couplers were classified in Class C (Figure 2). In this way, the proposed approach provides both a ranking of the candidate products and their classification into priority groups, which can support the product selection process.

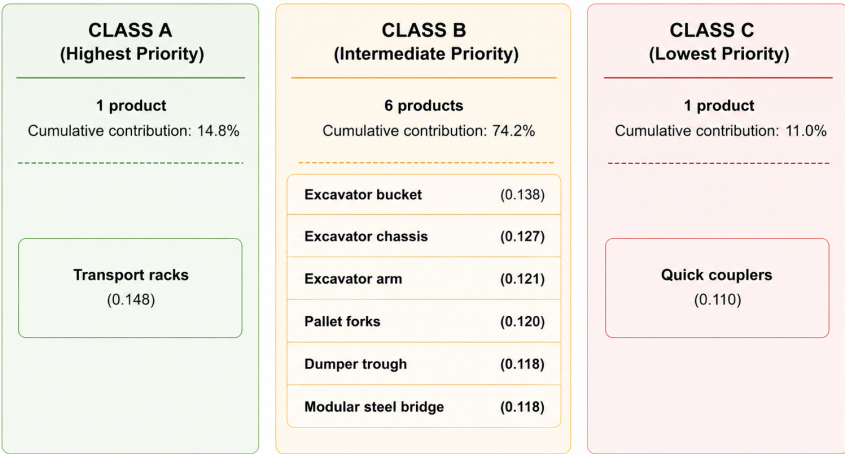


Figure 2. ABC classification of candidate products

Note: ABC refers to the Pareto classification of candidate products into Classes A, B, and C.

The results also show the effect of the dynamic criterion-weighting procedure used in the proposed fuzzy ABC approach. This procedure maintains the relative contributions of criteria within each ABC class and ensures that the final criterion weights follow the predefined ABC hierarchy. Consequently, criteria belonging to higher-priority classes cannot be assigned lower final weights than criteria belonging to lower-priority classes. This provides a more controlled distribution of criterion importance and prevents a criterion from a lower-priority class from receiving a disproportionately high weight simply because fewer criteria belong to that class.

Overall, the results demonstrate that the proposed fuzzy ABC approach can effectively integrate expert linguistic judgments, quantitative and qualitative product evaluations, dynamic criterion weighting, ranking and ABC classification within a unified decision-making framework. The approach therefore provides decision-makers with both a detailed ranking of candidate products and a higher-level prioritization structure that can be used to support decisions regarding the expansion of an existing manufacturing program.

5 Conclusions

This study developed a fuzzy ABC-based approach with dynamic criterion weighting for the prioritization and selection of new products for inclusion in an existing manufacturing program. The methodology combines linguistic expert judgments represented by TFNs, Pareto-based ABC classification, dynamic criterion weighting, and multi-criteria product evaluation. Unlike conventional ABC-based approaches, ABC classification is applied not only to the candidate products but also to the evaluation criteria, so that the class assigned to a criterion is taken into account when determining its final weight.

The main methodological contribution of the study is the dynamic scaling mechanism used to establish a relationship between the ABC classification of criteria and their final weights. The proposed procedure preserves the relative importance of criteria within each ABC class while ensuring that criteria belonging to higher-priority classes cannot receive lower final weights than criteria belonging to lower-priority classes. In this way, the information generated by the ABC classification is incorporated directly into the weighting process rather than being used only as a subsequent classification result. The mechanism therefore provides a consistent link between the relative contribution of criteria and their final decision weights.

The applicability of the proposed approach was demonstrated through a real-world case study involving the selection of new products for a manufacturing company producing welded steel structures and components for construction machinery. Nine experts evaluated ten criteria and eight candidate products using linguistic assessments for the subjective evaluation components. Transport racks achieved the highest overall decision value and was ranked first. It was followed by excavator bucket, excavator chassis, and excavator arm. According to the ABC classification, Transport racks were assigned to Class A, while six products were placed in Class B and quick couplers in Class C.

Thus, in addition to the ranking of all candidate products, the results also provide their grouping according to decision priority, which can be useful when considering the expansion of a manufacturing program.

From a practical perspective, the proposed approach can be applied when new product selection involves different types of criteria, expert knowledge, and potentially conflicting economic, market, technological, and production-related requirements. The use of linguistic assessments enables experts to express judgments without requiring artificially precise numerical estimates, while the combination of ranking and ABC classification provides two complementary views of the decision problem. The resulting framework can therefore support managers in distinguishing highly attractive products from alternatives requiring further analysis or lower-priority consideration.

Despite these contributions, the study has several limitations. First, the empirical validation is based on a single manufacturing company and a specific set of eight candidate products and ten evaluation criteria. Therefore, the obtained results should not be generalized directly to other industrial sectors or decision contexts. Second, the proposed weighting mechanism depends on the initial expert assessments and on the adopted Pareto thresholds for defining Classes A, B, and C. Changes in the expert panel, linguistic assessments, or classification thresholds may consequently influence the resulting criterion weights and product priorities. Finally, although the case study demonstrates the practical applicability of the approach, further comparative validation against other weighting and MCDM methods would provide additional evidence regarding its robustness.

Future research could therefore focus on applying the proposed fuzzy ABC approach to a larger number of industrial cases and different types of product-selection problems. Further research could also investigate alternative Pareto thresholds, different fuzzy representations, and alternative mechanisms for dynamic adjustment of criterion weights. The stability of the obtained rankings could also be examined under different evaluation conditions and parameter settings. Such analyses could provide additional insight into the robustness of the proposed approach.

Overall, the proposed approach combines fuzzy expert evaluation, ABC classification, and dynamic criterion weighting in the new product selection process. ABC analysis is used not only to classify the results but also as part of the criterion-weighting procedure. In this way, the contribution of a criterion and the class to which it belongs are reflected in its final weight. This represents the main methodological contribution of the study and provides a basis for further development of ABC-based MCDM approaches in complex industrial decision-making environments.

Data Availability

The data supporting our research results are included within the article.

Conflicts of Interest

The author declares no conflicts of interest.

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